

Postdoctoral Researcher Position

Deep Learning for Functional-Oxide Growth

Video-to-Spectrum Prediction by RHEED / XRD Fusion

Project:	OXYD-IA – Deep video-to-spectrum learning for the structural prediction of functional oxides (RHEED / XRD fusion).
Host labs:	INL (<i>Functional Materials & Nanostructures</i> team) & LIRIS (<i>Imagine</i> team), École Centrale de Lyon.
Location:	Écully campus (Greater Lyon), France.
Supervisors:	Clarisse Furgeaud (CNRS Researcher, INL) & Stéphane Derrode (Professor, LIRIS).
Funding:	École Centrale de Lyon (<i>Contrats Postdoctoraux 2026–2027</i>).
Contract:	Fixed-term public-law contract (<i>CDD de droit public</i>) with École Centrale de Lyon, full-time (100%), subject to confirmation of funding.
Duration:	1 year (12 months); approximate start around January 2027 (negotiable).
Application deadline:	October 30, 2026

Labs: INL, LIRIS, École Centrale de Lyon.

Keywords: deep learning, video understanding, uncertainty quantification, MBE, RHEED, XRD, functional oxides.

PROJECT OVERVIEW

We are looking for a highly motivated Postdoctoral Researcher to develop innovative deep-learning models that predict the structure of functional-oxide thin films directly from their growth dynamics. Positioned at the interface between **Artificial Intelligence** and **materials physics**, the **OXYD-IA** project aims to design a deep model able to predict the final **X-ray Diffraction (XRD)** spectrum of an oxide thin film from the sole **Reflection High-Energy Electron Diffraction (RHEED)** video recorded during its growth by **Molecular Beam Epitaxy (MBE)**.

The resulting tool will open the way to **predictive, in situ control** of oxide epitaxy – a process today dominated by a costly trial-and-error approach, in which the structural and functional properties of the films are only validated *ex situ*. You will join a genuinely multidisciplinary collaboration between the **INL** (*Institut des Nanotechnologies de Lyon*), which provides the *operando* experimental data and materials-physics expertise, and the **LIRIS** (*équipe Imagine*), which provides the deep video-learning and probabilistic-modelling methodology – both at École Centrale de Lyon.

A high-impact opportunity. AI for experimental physics is a fast-growing field, and OXYD-IA offers a genuine first-mover advantage within it: you would work on a **unique, unpublished dataset** of paired RHEED videos and XRD spectra to build one of the first video-to-spectrum models with **calibrated uncertainty** for oxide growth. By learning directly from experimental

data, such a model can short-circuit the traditional trial-and-error loop, **accelerate discovery** and drastically cut experimental time, cost, precursor consumption and instrument occupancy – turning routine *in situ* diagnostics into predictive tools.

Scientific context. Epitaxial perovskite-oxide thin films exhibit rich, tunable functional properties (thermoelectricity, ferroelectricity, piezoelectricity) governed by their structure and composition, themselves correlated with the growth conditions. MBE offers independent control of each element but, applied to oxides, is notoriously unstable and poorly reproducible because of the oxidising atmosphere. RHEED tracks the surface dynamics in real time (lattice parameter, surface reconstructions, roughness) but does not give access to the final *bulk* microstructure, which is only reachable *ex situ* via XRD / XRR. There is therefore a strong physical link – so far unexploited quantitatively – between RHEED dynamics and final XRD microstructure, which OXYD-IA proposes to **learn directly**.

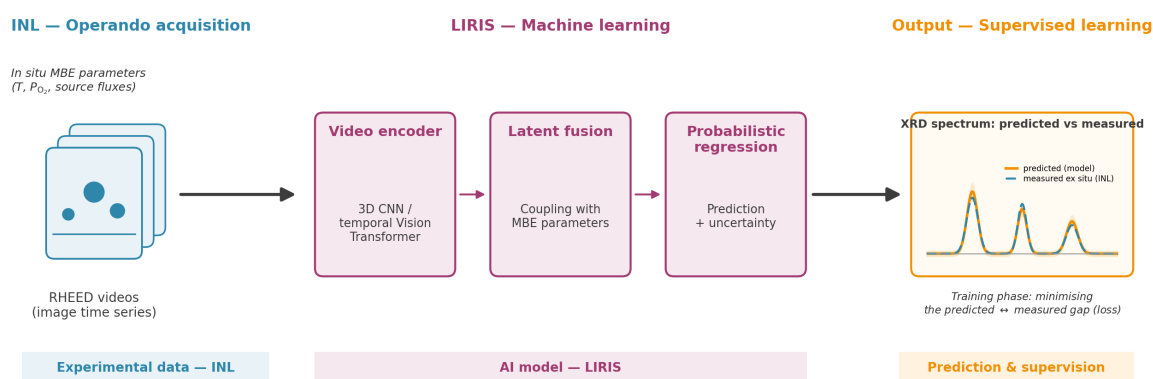


Figure 1 – OXYD-IA pipeline. At inference, the RHEED video acquired during growth (INL) feeds a deep video encoder (LIRIS) and a probabilistic regression module that predicts the final XRD spectrum together with its uncertainty. During training, the predicted spectrum is compared with the *ex situ* XRD / XRR spectrum measured at INL (ground truth); this discrepancy is the supervised learning signal that adjusts the model parameters.

KEY RESEARCH AREAS AND RESPONSIBILITIES

You will drive the core computational research along a risk-managed, phased 12-month roadmap. The **primary target** is a versioned RHEED / XRD dataset, growth-regime classification and probabilistic regression of the key structural parameters with calibrated uncertainty; full-spectrum prediction, a real-time demonstrator and cross-oxide generalisation are **stretch and follow-on objectives**. The work is organised around three technical pillars:

1. Corpus structuring and spatio-temporal RHEED representation

- Build a versioned dataset by cleaning, temporally aligning and pairing the RHEED video sequences, the associated MBE parameters (temperature, oxygen pressure, source fluxes) and the *ex situ* XRD / XRR spectra (SrTiO₃ reference system).
- Pre-train, in a self-supervised manner, a spatio-temporal video encoder (3D CNN, e.g. R(2+1)D, or temporal Vision Transformer) fine-tuned on the laboratory corpus.
- Perform fuzzy / Bayesian classification of the growth regimes (2D vs. 3D modes, transitions, RHEED oscillations), building on LIRIS expertise in the statistical analysis of sequences.

2. Probabilistic RHEED → XRD regression with uncertainty quantification

- Design a latent representation that couples the video encoding with the physical MBE parameters.
- Learn stepwise – first scalar structural parameters (lattice parameter, density, mosaicity), more robust at small N , then the full 1D XRD spectrum.
- Provide **calibrated uncertainty estimates** (Bayesian approaches or deep ensembles), indispensable for guiding real-time decisions during growth.

3. Real-time demonstrator and dissemination (*stretch / follow-on*)

- Validate the model on new SrTiO₃ growths and prototype an in-growth inference tool at the end of the NANOFUTUR MBE line.
- Mitigate the small-data and generalisation challenges through temporal data augmentation, transfer learning from pre-trained video encoders and domain adaptation towards other functional oxides.
- Contribute toward a high-impact joint publication at the AI / materials interface (the project milestone is the submission of a joint paper).

REQUIRED QUALIFICATIONS

- **A Ph.D. in the field of Artificial Intelligence** – Computer Science, Machine Learning, Computer Vision or Signal Processing, and we are looking for a senior postdoc with 2–5 years of post-PhD experience.
- **Deep learning on sequential / video data:** demonstrable, hands-on experience with 3D CNNs, temporal transformers and / or probabilistic models. Proficiency in **Python** and **PyTorch** (or TensorFlow) is mandatory.
- **A genuine appetite for physics:** a strong curiosity for, or prior exposure to, experimental physics and materials science. The ideal candidate carried out an AI-focused Ph.D. and is eager to apply it to a real materials-physics problem.
- **Research mindset:** a track record of innovation and the ability to work autonomously on a complex, multidisciplinary project.
- **Software skills:** experience with Git version control and reproducible research.
- Excellent communication and scientific-writing skills in English.

ADDITIONAL SKILLS AND COMPETENCIES

- Self-supervised video pre-training, uncertainty quantification (Bayesian / ensembles), fuzzy classification, or domain adaptation.
- Experience with small-sample / transfer-learning settings and handling large-scale experimental datasets.

WHY JOIN US

- A **12-month full-time postdoctoral position** at the AI / experimental-physics interface – a rare, career-structuring experience.
- **Bi-laboratory supervision** (INL & LIRIS) and access to **world-class MBE equipment** (Équipex NANOFUTUR) with real-time *in situ* diagnostics.
- A **unique, unpublished experimental dataset** and a high-potential application area for AI methods, with strong prospects for publication.

- A monthly gross salary of **2,800 euros**.
- French **social-security coverage** (health, pension, unemployment) and statutory paid annual leave.
- Assistance with **relocation and visa / residence-permit** procedures for international (non-EU) candidates.
- Access to high-performance GPU computing resources, in a dynamic and collaborative research environment.
- Alignment with the *AI for engineering and materials science* strategy of École Centrale de Lyon, and involvement in follow-up collaborative proposals (ANR PRC, PEPR DIADEM, PEPR Électronique).

HOW TO APPLY

Please apply to the position on the website: <https://ecolecentraledelyon.recruitee.com/o/centrale-lyon-postdoctoral-researcher-position-deep-learning-for-functional-oxide-growth-video-to-spectrum-prediction-by-rheed-xrd-fusion>

Submit your application before **October 30, 2026**. Informal enquiries are warmly welcomed by email to either supervisor before applying.

Selection process. Shortlisted candidates will be contacted within about three weeks of the deadline for an interview (online or on-site) with both supervisors; the final decision is expected by the end of 2026, for a start negotiable around January 2027.

Data protection (GDPR): your application is processed solely for this recruitment, is accessible only to the selection panel, and is kept no longer than the process requires; you may request access to or erasure of your data at any time. Please obtain your referees' consent before sharing their contact details.

École Centrale de Lyon is an equal-opportunity employer and welcomes applications from all qualified candidates regardless of gender, origin, disability or any other protected characteristic.